

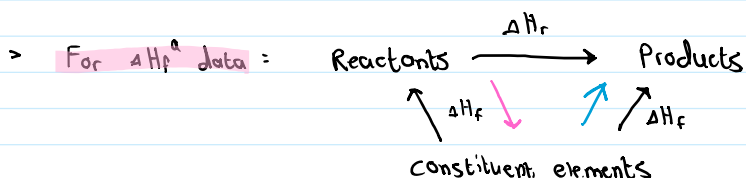
Enthalpy change of formation:

the change in energy when one mole of a compound is formed from its constituent elements in standard states and under standard conditions.

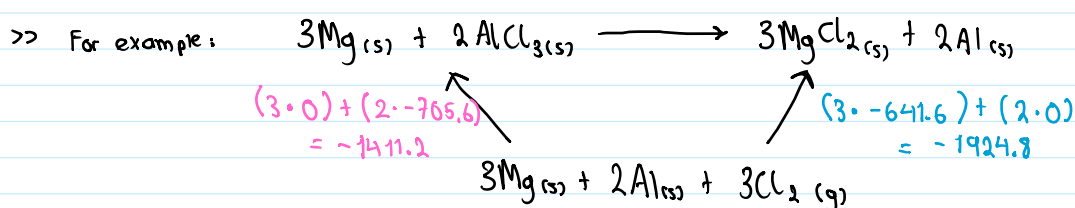
Enthalpy change of combustion:

the energy change when one mole of a compound is completely combusted in standard states under standard conditions with excess oxygen.

Hess' Law

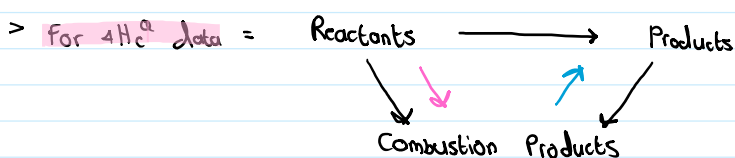


>> Furthermore, $\Delta H_r^\circ = \sum \Delta H_f^\circ(\text{Products}) - \sum \Delta H_f^\circ(\text{Reactants})$

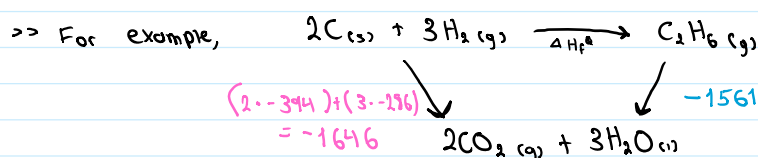


Formation Data	Mg(s)	AlCl ₃ (s)	MgCl ₂ (s)	Al(s)
ΔH_f° (kJmol ⁻¹)	0	-705.6	-641.6	0

Finally, $\Delta H_r^\circ = -1924.8 - -1411.2$
 $\therefore \Delta H_r^\circ = -513.6 \text{ kJmol}^{-1}$



>> Furthermore, $\Delta H_r^\circ = \sum \Delta H_c^\circ(\text{Reactants}) - \sum \Delta H_c^\circ(\text{Products})$



Combustion data	C(s)	H ₂ (g)	C ₂ H ₆ (g)
ΔH_c° (kJmol ⁻¹)	-394	-286	-1561

Finally, $\Delta H_r^\circ = -1646 - -1561$
 $\therefore \Delta H_r^\circ = -85 \text{ kJmol}^{-1}$

> Sometimes $\Delta H_r^\circ = \Delta H_f^\circ$

>> You may be given ΔH_c° data and be asked to calculate ΔH_f° rather than ΔH_r°

>> In this instance, use the same method as above (ΔH_c° data) and your answer will be ΔH_f° rather than ΔH_r°